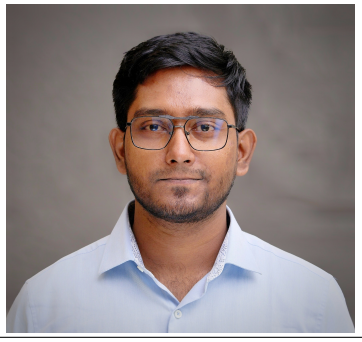
# Mushfiqul Islam Chowdhury

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## Introduction

M.Sc. Artificial Intelligence student at the University of Jyväskylä with 4+ years of experience in applied AI, software engineering, and data-driven system development. Experienced in building production-grade AI systems including multi-agent architectures, RAG pipelines, conversational AI interfaces, LLM-powered applications, and ML-enabled services. Strong in Python, backend engineering, and production-ready AI workflows. Hands-on with CI/CD using GitHub Actions and Docker Compose. Published IEEE conference work on real-time action recognition, with experience in PyTorch, TensorFlow, and reproducible AI workflows.

## Education

**University of Jyväskylä**, M.Sc. in Artificial Intelligence — Aug 2025 – Present

- Coursework and projects emphasize empirical ML evaluation, data-driven system design, and responsible AI practices.
- Focus areas: Applied machine learning, NLP/LLM applications, and practical AI systems.

**BRAC University**, B.Sc. in Computer Science and Engineering — Jan 2017 – Sept 2021

- Undergraduate Thesis: Real Time Action Recognition from Video Footage

## Professional Experience

**Software Engineer**, Pipeline Bangladesh Ltd — May 2022 – Aug 2025

- Architected and led Risk Sensor (risksensor.io), a production multilingual AI security analysis platform using ReAct-style multi-agent orchestration and RAG pipelines, reducing a one-week manual review process to near-real-time automation.
- Translated complex security workflows into scalable AI-enabled backend services, collaborating with stakeholders from requirements through deployment, code review, performance tuning, failure handling, and operational handover.
- Designed and optimized high-throughput data pipelines processing 120K+ events/logs per minute, reducing memory consumption by over 95% while improving throughput and deployment efficiency.
- Applied production AI practices including structured evaluation, reproducible workflows, observability considerations, and fallback-aware system design.
- Collaborated on web-based interfaces and internal tools for AI-assisted review, monitoring, and workflow management.

**Intern**, Kimberton AI Studio — January 2026 – April 2026

- Contributed to the design and implementation of automation pipelines and AI-assisted workflows, with a focus on backend logic, orchestration, and system reliability.
- Worked on web-based AI application components using React/Next.js and API-driven backend integration.

**Junior Software Engineer**, Tirzok Private Limited — June 2021 – Apr 2022

- Designed and implemented a graph-based reward engine using Neo4j and custom Odoo modules to

automate rank calculation and reward distribution for e-commerce transactions.

- Acted as a coordination point between backend engineering and product design teams to align implementation with product intent.

**Junior Software Engineer**, Part-time, Field Information Solutions Ltd (Field Buzz) Apr 2020 – May 2021

- Redesigned role-based access control by modularizing permissions, improving usability and security.

## Selected Projects

**Sales Chatbot** Python, LangChain, OpenAI API, PostgreSQL sales-chatbot

- Built a generative AI chatbot for personalized product recommendations and order tracking using behavioral and country-level buying data.
- Designed a layered architecture separating AI orchestration, business logic, retrieval, and data access.

## Leadership & Community

**General Secretary**, BRAC University Computer Club (BUCC) Jul 2019 – Jul 2020

- Organized programming contests, peer-learning workshops, and technical training sessions for students.
- Collaborated with faculty, industry partners, and student clubs to promote technology awareness and campus engagement.

**Speaker**, Google Developers Group (GDG) Dhaka Sep 2024

- Delivered a talk on "Career with AI," sharing practical guidance on responsible and applied AI.

## Publications

**Real-Time Action Recognition from Video Footage** 2021

Mushfiqul Islam Chowdhury, Tasnim Sakib Apon, Md Zubair Reza, et al.
IEEE conference publication.
DOI: 10.1109/STI53101.2021.9732601

## Technologies

**Programming:** Python (Primary), SQL (Strong), Go (Strong), JavaScript (Working), TypeScript (Working), Bash (Working)

**AI/ML & Data Science:** Claude Code, Cursor, MCP, LLM applications, RAG, ReAct agents, LangChain/LangGraph, OpenAI API, Anthropic API, NumPy, Pandas, scikit-learn, TensorFlow, PyTorch

**Frontend & Full-Stack:** REST APIs, FastAPI, React, web-based AI applications, API integration

**Cloud / MLOps:** AWS (EC2, ECS, SageMaker, Bedrock), GCP (Firestore, Vertex AI), Docker, Kubernetes, GitHub Actions (CI/CD), MLflow, ZenML

**Testing & Quality:** TDD, structured AI evaluation

**Data & Backend:** PostgreSQL, Kafka, Apache Spark, Redis, Qdrant, Neo4j, Django, DynamoDB, SQLite

## Languages

English (Fluent), Bengali (Native), Finnish (Basic, actively improving)